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RESEARCH

Automated geolocalization of vehicles from UAV footage: evaluating measurement precision of object detection and segmentation methods

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Abstract

Modern Road Traffic Monitoring (RTM) systems rely on advanced and precise technologies. Unmanned Aerial Vehicles (UAVs) coupled with state-of-the-art computer vision methods offer great utility in intelligent traffic monitoring and road safety analysis. However, the precision of these cutting-edge technologies is still under debate due to technical complexities, such as inaccurate road-user localization resulting in overestimated bounding dimensions, which could hinder their effectiveness in real-world scenarios. This research introduces a geolocalization method combining a feature-matching algorithm, SIFT, for automatic georeferencing of UAV frames with deep learning-based object detection and segmentation models. The study focuses on finding the most precise solution for vehicle geolocalization, preserving the vehicle shape and dimensions. The study explores three different configurations of YOLO object detectors: a standard YOLOv8 model, a hybrid model that integrates YOLOv8 with the Segment Anything Model (SAM), and a YOLOv8 variant that employs Oriented Bounding Boxes (OBB). The evaluation of results is focused on the dimensional accuracy, internal variabilities, impact of altitude variations, vehicle tilt or rotation, and inference speed of each method. Experimental results reveal that the YOLOv8 coupled with SAM and the YOLOv8-OBB exhibit comparable precision and excel in accurately localizing road users while preserving their dimensions. This can be instrumental in a practically feasible vision-based RTM solution. In terms of speed-to-error ratio, OBB-enabled object detectors present the most practical option, allowing for near-real-time solutions in key road safety workflows, such as conflict analysis.

Keywords UAV · Geolocalization · Measurement · Road safety · Deep learning · YOLO

Introduction

Given the global surge in vehicular traffic, conventional road traffic monitoring (RTM) methods are gradually proving ineffective. This emerging challenge demands the adoption of Intelligent Transportation Systems (ITS), which seek to mitigate traffic management issues through the integration of state-of-the-art technologies, notably Unmanned Aerial Vehicles (UAVs) and Artificial Intelligence (AI)-based object detection models. The utilization of UAVs marks a paradigm shift, offering an aerial perspective that is non-existent in other methods. UAVs had transformative impacts across various fields of science, including agriculture, earth observation, geology, and climatology (Tewes 2018).

The popularization of AI has enabled the evolution of smart cities, positioning them at the forefront of global urban development strategies. RTMs are a critical infrastructure component of smart cities that ensure road-user safety and facilitate smooth traffic flow, thereby improving the quality of life for urban dwellers (Elloumi et al. 2018). Despite the remarkable progress over the past decade, much of the

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research remains fragmented across specific disciplines or narrowly defined tasks, often relying on controlled assumptions that limit direct applicability in complex real-world environments. In the transportation domain, advances in computer vision and deep learning (DL) have made it practically feasible to detect and localize road users from video data with a high degree of automation (Zhou et al. 2022; Duan et al. 2018). Adopting these vision-based DL models into automated workflows could reduce the need for manual intervention, thus optimizing traffic management operations. Nevertheless, deploying such state-of-the-art technologies comes with its inherent challenges. Estimating a vehicle's precise location and dimensions is crucial for practical road safety improvements. This precision in measuring distances between road users is essential for effectively implementing the surrogate safety indicators such as Time to Collision (TTC), Post-Encroachment Time (PET), Time Advantage, and Time Gap. This precision plays a central role in ensuring the integrity of these safety metrics (Jiang et al. 2021; Johnsson et al. 2018; Laureshyn et al. 2010). Current applications of AI in traffic safety research face several limitations that hinder large-scale adoption, particularly in complex, real-world traffic environments where multiple vehicles interact simultaneously. As vehicles change their orientation in motion, for example, during turning or lane changes, their bounding box sizes often fluctuate, impacting the positional accuracy and size estimates. Because surrogate safety measures such as PET and TTC rely on road users' actual positions, distances, and footprints, overestimated bounding boxes can result in flawed assessments of safety thresholds, as illustrated in Fig. 1. To develop an automated road safety solution, it is essential to review existing object localization methods in real-world experiments and identify the most accurate solution.

To address this knowledge gap, we present this research by conducting experiments on several state-of-the-art detection and segmentation models and evaluating their localization accuracy in real-world environments. The study introduces an automated solution for vehicle geolocalization from UAV footage, combining a feature-matching algorithm SIFT for georeferencing with three configurations of the 8th-generation YOLO object detector: a standard YOLOv8 model, a hybrid model that integrates YOLOv8 with the Segment Anything Model (SAM), and a YOLOv8 variant that employs Oriented Bounding Boxes (OBB). Georeferencing plays a critical role in transforming pixel-based YOLO detections into precise real-world positions and dimensions, which is essential for surrogate safety measures, as illustrated in Fig. 1. The accuracy of geolocalization was assessed by comparing bounding boxes from each YOLO variant with ground truth, considering precision, internal fluctuations, and sensitivity to factors such as UAV altitude and vehicle orientation. The results provide insights into the strengths and limitations of each approach and demonstrate the potential for implementing automated road safety workflows using object detection and segmentation models in real-world scenarios.

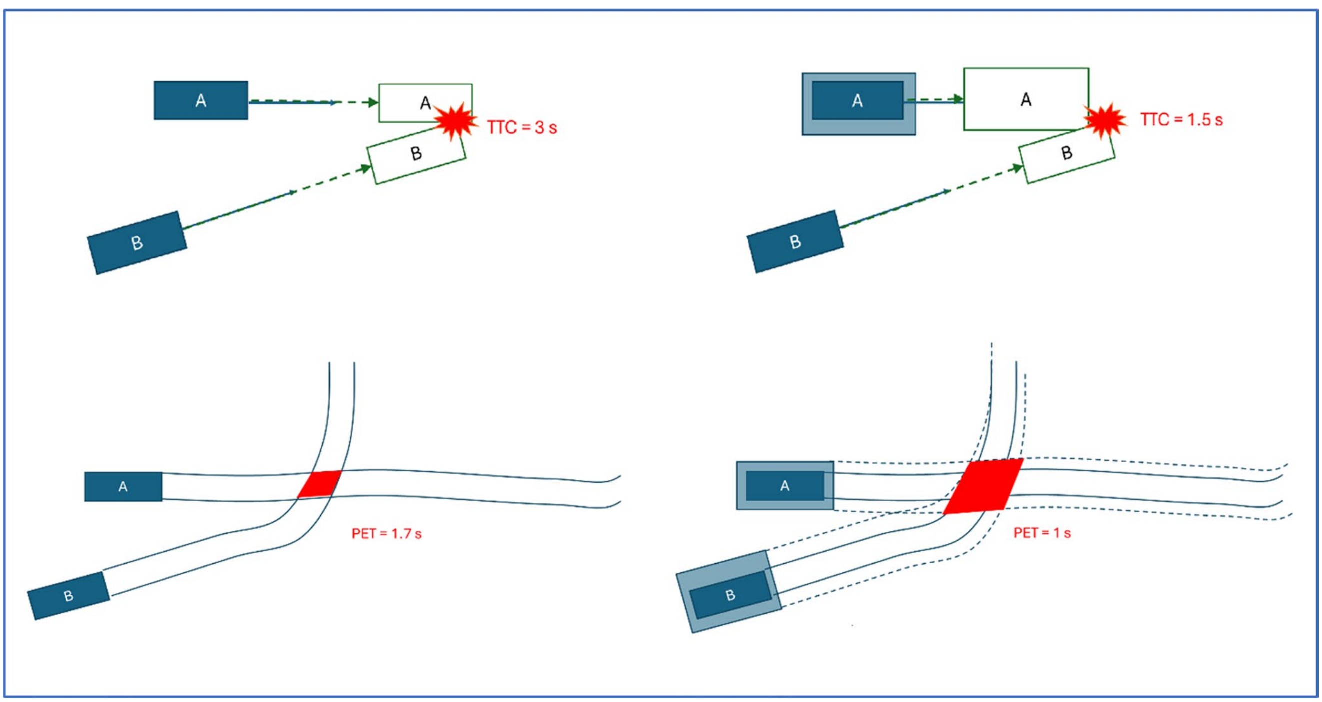

Fig. 1 An illustration of how overestimated vehicle bounding boxes could lead to wrong estimation of TTC and PET

Related works

With the recent technological advancements in AI, computer vision, and UAV-based systems, researchers have been exploring the potential of these cutting-edge technologies to enable automation within different domains (Chen et al. 2023). One prominent application is traffic monitoring and road safety, where accurate localization of vehicles is critical. A fundamental step toward object detection in real geographic space is the calibration of video pixels into real-world coordinates, a process known as georeferencing. In this process, pixel coordinates are converted into a geographical coordinate system. However, georeferencing of UAV-based data often poses a challenge due to the dynamic nature of UAV platforms. Ahmed et al. (2025) addressed this issue by proposing a feature-matching-based automated near-real-time solution. The authors employed the SIFT feature-matching algorithm for automated georeferencing and subsequently blurred private properties in a UAV livestream. Feature-matching algorithms are often used for visual odometry, principally for UAVs' absolute visual localization (AVL) in GPS-denied environments. The underlying principle behind these methods is to match the UAV's current visual input with a pre-existing satellite map or reference aerial imagery from an earlier flight (Couturier and Akhloufi 2021). Long et al. (2016) developed a method for correcting geometric biases in remotely sensed aerial imagery by using the Scale-Invariant Feature Transform (SIFT) algorithm coupled with least squares matching (LSM) to improve the accuracy. This method employed pre-existing online aerial imagery to align and match larger remote sensing (RS) images. The approach enabled precise and efficient image matching by segmenting a large image into smaller and manageable tiles. The results confirmed that the proposed methodology could identify ground control points (GCPs) in tens of seconds with satellite images having spatial resolution ranging between 30 m and 2 m, proving it to be a convenient solution for automated georeferencing.

The utilization of YOLO-based Convolutional Neural Network (CNN) architectures has increased, primarily due to their unparalleled inference capabilities and noteworthy accuracy. A significant contribution to this field was made by Tao et al. (2017), where the authors conducted a comparative analysis of various algorithms, including Fast-RCNN, Faster-RCNN, R-FCN, YOLO, SSD, OYOLO, and OYOLO + RFCN. This study employed a custom dataset derived from road cameras, integrated into the KITTI dataset, with a training-to-testing ratio of 9:1. The results indicated that Optimized YOLO (OYOLO) achieved the highest inference speed at 44 ms while maintaining a precision rate of 80.1%. The authors also proposed the integration of OYOLO with RFCN to enhance precision. Another research work by Zuraimi and Zaman (2021) explores the implementation of YOLOv4 with the DeepSORT algorithm for vehicle detection and tracking, on footage captured by roadside cameras. These authors benchmarked the combined performance of custom-trained YOLOv4 and DeepSORT against earlier YOLO architectures, revealing that their proposed method surpasses all previous algorithms, achieving a mean Average Precision (mAP) of 82.08%.

UAVs have significant advantages over roadside camera systems, primarily due to their enhanced mobility, extensive coverage, and exclusive aerial perspective. Integrating UAVs with advanced computer vision technologies can facilitate the development of a robust traffic monitoring system. Chen et al. (2023) thoroughly reviewed YBUT (YOLO-based UAV technology), tracing its technological evolution within engineering, transportation, and other civil applications. Researchers have also explored object detection in real geographical spaces, also known as geolocalization, which provides the foundation for our study. A notable contribution to this field is the work by Pratama et al. (2020), which introduces a novel approach utilizing a YOLO-based CNN architecture on aerial imagery. The model is trained on a dataset comprising Google Earth images and then applied to aerial imagery in Geotiff format to detect and localize three specific object classes: airplanes, cars, and ships. Notably, the system demonstrated an accuracy of 81.05% and a positional Mean Absolute Error (MAE) ranging between $0.000012^{o} - 0.000034^{o}$. In another research, Coulibaly et al. (2023) integrated geographical information derived from aerial imagery with advanced detection capabilities of YOLOv8 to geolocalize degraded asphaltic pavement. To achieve this, high-resolution aerial photographs captured by a UAV were employed to train the YOLOv8 model. The model accurately detected and localized degraded pavement sections and logged the geographical coordinates of these degraded segments. The experimental results revealed that the model exhibited a precision of 86.7%, a recall of 78.8%, and an F1 score of 82.5%. The findings also underline the model's efficiency in distinguishing between various forms of pavement degradation, including cracks, ruts, and potholes. The results of this study could be used to facilitate road maintenance activities, highlighting a practical use case for disruptive technologies in smart city and road safety applications.

In road traffic monitoring and road safety solutions, the robust inference capabilities of YOLO and the precise georeferencing abilities of feature-matching can be combined to develop fully automated UAV-based road traffic monitoring solutions. A recent study on vehicle geolocation using UAV footage combines a feature-matching algorithm for georeferencing with the deep learning model YOLOv8. Although the authors focused on the positional accuracy of vehicles, the dimensional accuracy of geolocalized vehicles is still a research gap that can be explored (Novikov et al. 2024).

Materials and methods

The feature-matching-based AVL methods employed for UAV localization in GPS-denied environments can be effectively adapted for geolocalizing road traffic. Feature-matching methods are generally free from error accumulation or drift, which GNSS-based and Relative Visual Localization (RVL) methods produce; therefore, the pixel calibrated through feature-matching is extremely precise (Couturier and Akhloufi 2021). The study proposes a method that adopts the principles of AVL to geolocalize vehicles while preserving their shape and dimensions. The method combines advanced object detection and segmentation techniques with a geospatial approach. As illustrated in Fig. 2, the initial step involves the creation of an orthomosaic, which serves as a reference image for feature matching and coordinate transformation processes.

This template or reference image is generated on a UTM-based projected coordinate system using an open-source tool named WebODM, a product of the OpenDroneMap™. For the experiment, two sets of footage were collected at differing altitudes (60 and 90 m) and locations during daylight hours under clear sky conditions using the DJI mini 3 pro at 3840 × 2160 resolution @ 30FPS. The experimental footage was recorded in daylight hours for the feature-matching algorithm to function. A Python script was developed and tested on a workstation with a 3.60 GHz Intel® Core™ i9-9900 processor and 64GB RAM to facilitate the accurate positioning of objects. Feature matching with georeferenced aerial imagery was used to compute a homography matrix, producing a planar projective transformation that converts pixel coordinates into the corresponding UTM coordinates. This transformation enables object detection in real-world space, allowing for analysis of vehicle dimensions in metric units. After the calibration process, three different algorithmic settings of YOLOv8 were used to detect, track, and localize vehicles in the sample frames. The first setting employed the conventional YOLOv8 model. The second setting combined YOLOv8 with the Segment Anything Model (SAM) developed by Meta AI. This method applies a segmentation mask to the detected object to refine the bounding box dimensions. The third setting used an oriented bounding box (OBB) enabled version of YOLOv8. These three configurations were chosen as the study focuses on the dimensional accuracy of detected objects instead of detection accuracy, which has been extensively researched and established previously. The resulting detections consist of four bounding box corners in UTM projection for an object across all sample frames. These bounding boxes were then compared against ground-truth measurements, which

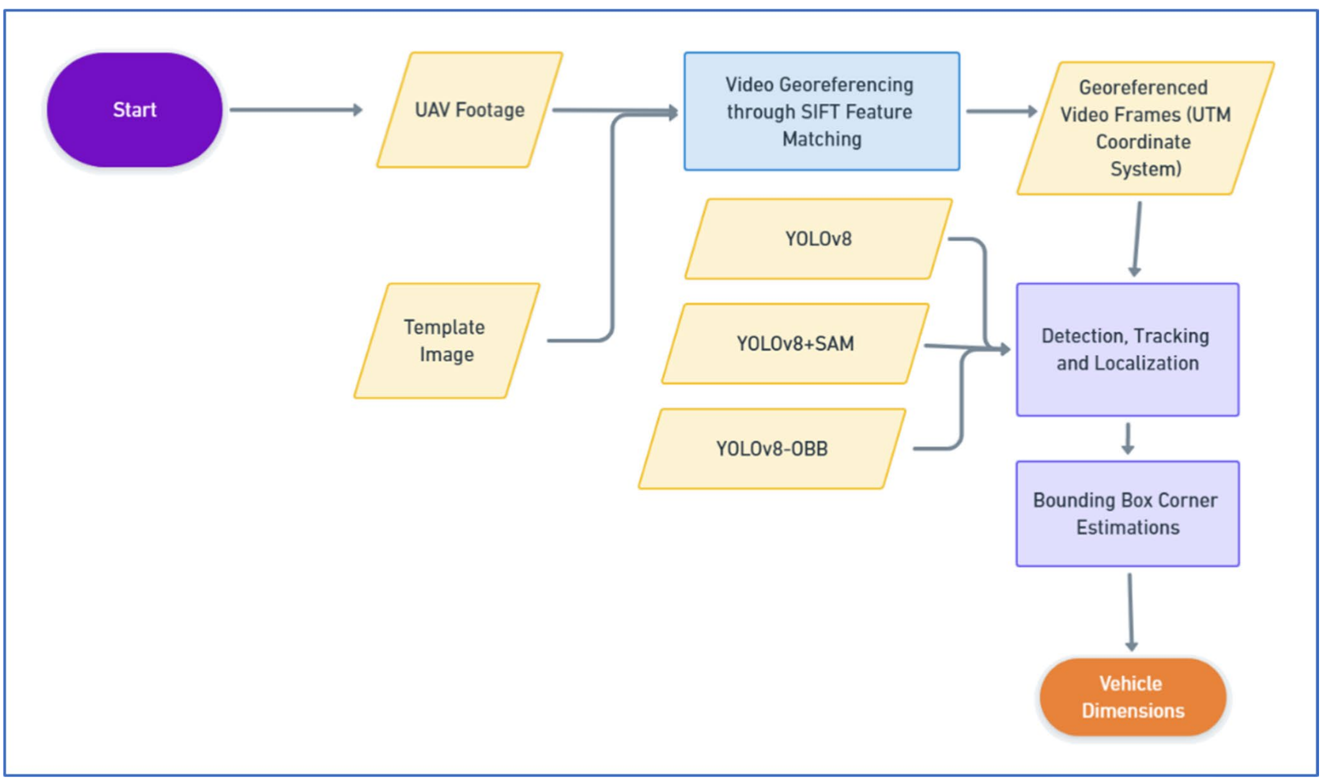

Fig. 2 Methodological framework of the study

were obtained from the reference image using the measurement tool in QGIS. All three methods were evaluated based on both accuracy and inference speed.

Calibration of UAV frames in projected coordinate system using SIFT

To enable spatial awareness within video data, it is essential to convert pixel coordinates into a real-world coordinate system. This enables the estimation of important traffic flow parameters, as it requires the physical location of vehicles across time (Ahmed et al. 2024). When combined with existing reference maps, feature-matching algorithms can automate this calibration process for UAV-based traffic footage (Ahmed et al. 2025). In this study, the pixel calibration process was automated by employing an automated georeferencing workflow using the feature-matching algorithm SIFT developed by Lowe (2004). The SIFT process can be divided into four stages. The first stage is scale-space extrema detection, where a Gaussian scale-space pyramid is created. This involves generating successive blurred images through convolution with Gaussian functions, thus forming multiple octaves. The parameter sigma establishes the initial Gaussian blur for the first level of each octave. In this stage, the Difference of Gaussian (DoG) method is applied, calculating the difference between images at different scales within an octave. The parameter n-scale indicates the number of scales sampled in each octave. While increasing the number of scales can result in more detected key points, it may also decrease their stability. A balance must be struck between the total number of key points and the stable key points found. The second step is feature localization, where a detailed fitting is conducted for each extremum to ascertain its location, scale, and the ratio of principal curvatures. Keypoints found in areas of low contrast or along edges are filtered out to ensure the robustness of the feature set. Next is the orientation assignment phase, where an orientation is assigned to each key point based on the local image gradients. This step is crucial for achieving rotational invariance in the final feature descriptors. Finally, during the feature descriptor creation stage, SIFT computes a descriptor for each key point invariant to residual variations. Each descriptor is a vector comprising 128 elements, which encapsulates pixel properties within the vicinity of a feature (Castillo-Carrión and Guerrero-Ginel 2017). SIFT has greater tolerance to seasonality due to its ability to match features despite variations in pixel intensities caused by variations in illumination or other seasonal causes, but the computational time for SIFT is exceptionally high, making it unsuitable for near real-time applications (Karami et al. 2017); (Van Quan et al. 2023). Speed optimization methods can be implemented to address the computational complexity of SIFT, including Principal Component Analysis (PCA) to reduce the dimensions of vector descriptors and limit the key-point area. These adjustments can make SIFT more effective for visual odometry and similar applications (Van Quan et al. 2023).

The preconstructed orthomosaic was used as a reference image to construct a homography matrix using SIFT descriptors and convert pixel coordinates of input footage to UTM coordinates. To develop this image, a DJI Mini 3 Pro drone was flown at a height of 90 m, equipped with a 48-MP camera set at a top-down orientation. Multiple images were captured by maintaining a visual line of sight (VLOS) during the data acquisition flight, and an orthomosaic with a UTM projection system was generated using the WebODM application of OpenDroneMap™. This orthomosaic image served as a reference image for the feature-matching algorithm to match and calibrate input frames and produce georeferenced output frames in the UTM system (see Fig. 3).

After matching key points from the input frame with the reference image and descriptor generation, a Lowe's ratio test was used to filter out the unsuitable matches, and the Random Sample Consensus (RANSAC) method was used to further filter out the outliers (Kaplan et al. 2016). Based on the homography matrix obtained from the best matches, pixel coordinates of input frames were converted to corresponding real-world coordinates. This facilitates the

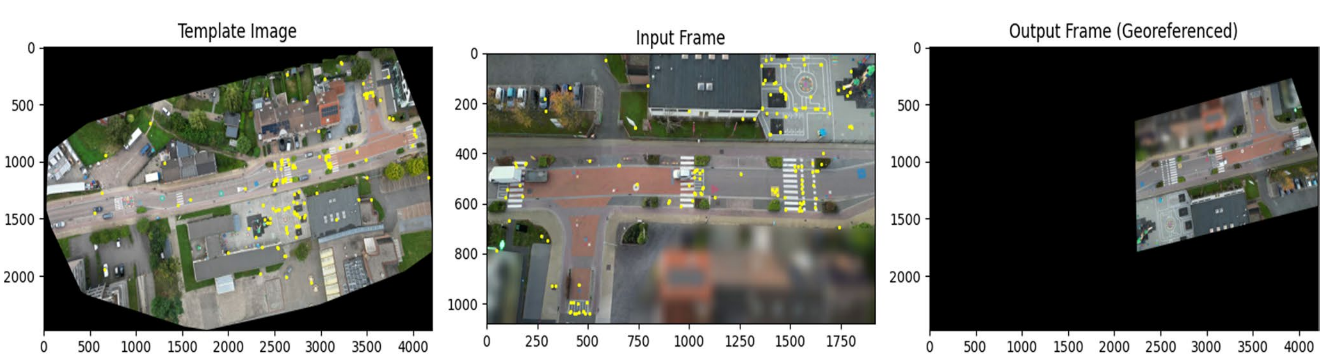

Fig. 3 The SIFT-based pixel-to-geographical coordinate system conversion of the input frame after feature-matching with the template image. The output is a georeferenced image with UTM-based coordinates

geolocalization of cars in the input image, converting the bounding box corner coordinates into the UTM coordinate system and enabling real-world measurements. The method is designed for two-dimensional frames; therefore, the camera angle of the UAV during footage acquisition must be identical to that of the reference image (in this case, 90 degrees or top-down) for the feature-matching algorithm to function correctly.

Vehicle detection and localization

8th generation you only look once (YOLO) architecture

YOLO algorithm represents a significant jump forward in object detection, introducing a single-stage detection. YOLO divides the input image into a grid, simultaneously predicting bounding boxes and class probabilities. This grid-based approach allows for faster inference, enabling real-time applications (Boudjit and Ramzan 2022). YOLO algorithm has seen numerous enhancements over the years, resulting in several versions, each offering improvements in detection precision and inference speed. The eighth generation of YOLO architecture has drawn considerable attention due to its superior performance in object detection compared to its predecessors and in some cases its successors (Sundaresan Geetha et al. 2024). Researchers have rigorously evaluated YOLOv8 in diverse experimental setups. The proposed method employed YOLOv8 for detection and its default tracking algorithm, Bot-SORT. The Bot-SORT algorithm possesses the ability to reidentify objects even after they have momentarily vanished from view, ensuring seamless and accurate tracking, which is a critical feature for applications demanding continuous monitoring of objects over time, such as RTM (Kalake et al. 2021; Yang et al. 2024). The rationale for using YOLOv8 instead of its successors was its superior performance for detecting larger road user classes, such as cars, vans, and trucks. In contrast, the 10th-generation YOLO architecture exhibits better detection capabilities for smaller objects, such as pedestrians (Sundaresan Geetha et al. 2024).

The selected YOLOv8 weights were trained on the VisDrone2019-DET dataset – specifically designed for object detection of low-lying objects (Chen et al. 2021). The model exhibited an average precision AP @0.5 of 64% for the class of interest: cars, which is adequate, as all the vehicles in the experiment footage remained detected throughout all sample frames and were manually verified for each sample frame. This combination of YOLOv8 and Bot-SORT enabled the consistent localization of each vehicle's object with its ID throughout the sample frames. The bounding boxes corresponding to each track were geolocalized within the UTM coordinate system. This was accomplished by converting pixel coordinates into their corresponding UTM coordinates, an approach adopted and tested in an earlier work for vehicle trajectory localization (Ahmed et al. 2024). The coordinates of all four corners of the bounding box were logged into GeoJSON format and mapped on the QGIS platform for analysis.

Segment anything model (SAM)

Precise geolocalization is essential for estimating vehicle headway and calculating other road safety measures. SAM by Meta AI has been a significant milestone in this domain, presenting a model with unique segmentation capabilities for all kinds of visual inputs (Kirillov et al. 2023). This functionality distinguishes SAM from the other models, positioning it as one of the foundational segmentation models in computer vision. SAM has been trained on an SA-1B dataset comprising 11 million images and 1 billion masks. SAM has shown proficiency in generalizing across various images and objects, emphasizing its potential to transform image analysis and understanding, which has been established in earlier research works (Deng et al. 2023; Ji et al. 2023; Li et al. 2024; Mazurowski et al. 2023).

To enable segmentation capabilities within the YOLOv8 model without necessitating further training, a hybrid model was developed by integrating YOLOv8 with SAM (Khatua et al. 2024; Zhang et al. 2024). This approach leverages the same YOLOv8 model, trained on the Visdrone dataset. Once a vehicle is detected, the generated bounding boxes are used as inputs for SAM, which allows the extraction of object masks and contours. Subsequently, the largest contour in each dimension (x and y) was selected as a benchmark to refine the bounding box dimensions for each vehicle (see Fig. 4), thereby minimizing the overestimation of the area covered. Additionally, the estimation of object contours facilitates the rotational refinement of the bounding boxes, which the conventional YOLOv8 fails to provide.

YOLOv8 oriented bounding boxes

Recent YOLO architectures (YOLOv8 onwards) offer a new feature called oriented bounding boxes (OBB), capturing the orientation of detected objects. Earlier YOLO versions utilized axis-aligned bounding boxes (AABB), lacking angular adjustments and, therefore, do not capture the true orientation of detected objects. The YOLOv8-OBB model trained on the DOTA dataset incorporates an additional parameter—the object's angle—to locate objects precisely (Feng et al. 2024).

YOLOv8 OBB model maintains the existing attributes of its predecessors while addressing the limitations of conventional bounding boxes (see Fig. 5). The OBB variant enables

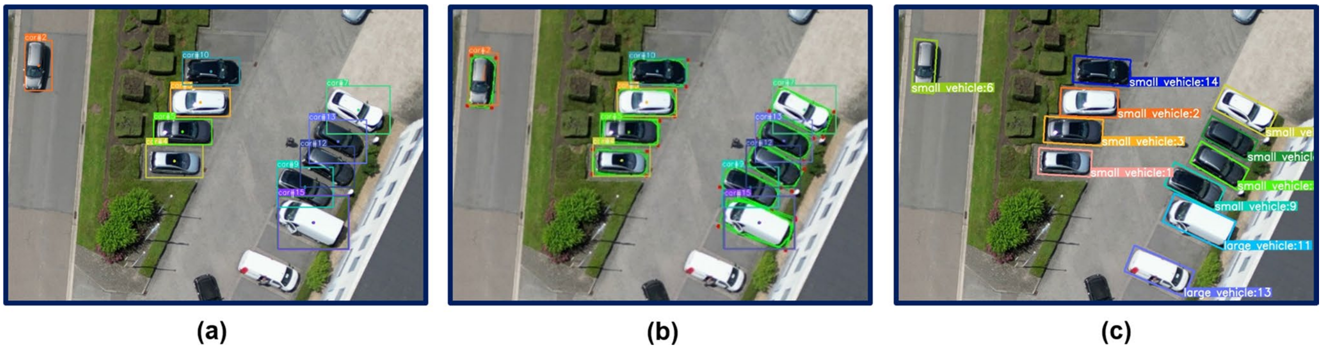

Fig. 4 A step-by-step reconstruction of rotated bounding boxes using YOLOv8 + SAM models. (**a**) shows the original detection using YOLOv8, (**b**) demonstrates segmentation masks implemented by SAM, (**c**) displays the extraction of contours from segmentation masks, and (d) visualizes the corrected bounding boxes estimated from the minimum area rectangle enclosing the object's contours

Fig. 5 A side-by-side comparison of (**a**) YOLOv8 with the (**b**) new YOLOv8 variant with OBB-enabled localization

Table 1 Summary of experimental footage

Data	Altitude (m)	Unique Tracks	Sampled Frames	Bounding Boxes
Footage 01	60	17	300	5,100
Footage 02	90	73	300	21,900

improved localization of objects. It avoids overestimation of object bounds, which can be of extreme advantage for road safety applications where dimensions and distances are important.

Results and discussion

Experimental findings

The experimental footages were collected in Hasselt, Belgium, at two distinct locations, flown at altitude levels of 60 and 90 meters, respectively. This variation in altitude was selected to evaluate the impact of flight altitude on vehicle localization accuracy. 300 sample frames were extracted from the two experimental videos (as discussed in Table 1) for the analysis. The first footage, captured at 60 m, included 17 unique vehicle tracks, yielding approximately 5,100 localized bounding boxes. The second footage, recorded at 90 m, contained 73 distinct vehicle tracks, with object detection producing 21,900 bounding boxes, all in the UTM coordinate system.

For ground truthing, the dimensions or footprint of the vehicle in the reference image were manually measured in square meters using the measurement tool in QGIS. These reference measurements were then used to calculate the RMSE for each track by comparing the estimated vehicle footprint from the detected bounding boxes with the manually measured values (see Eq. 1). Manual tracing was preferred over the manufacturer's dimensions due to the problem's visual nature. Factors such as optical occlusions, shifting shadows, and overall errors in the reference image can influence the representation of the detected

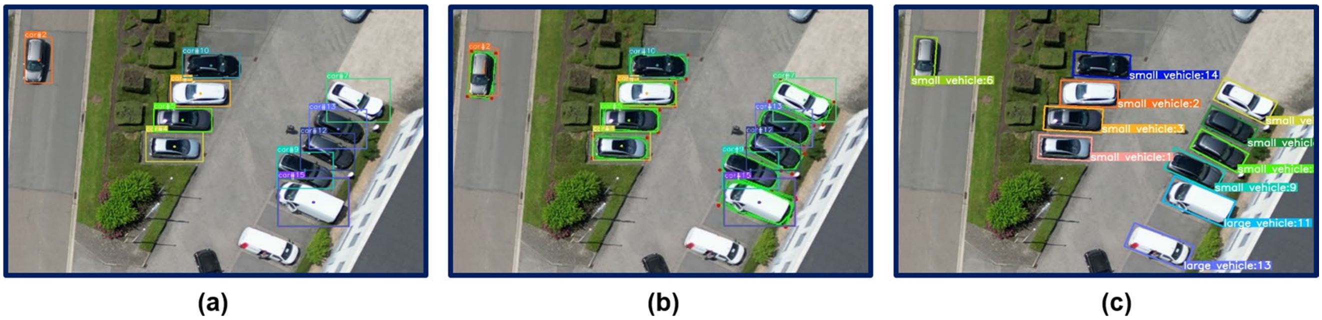

Fig. 6 Geolocalization of vehicles from the three methods: (**a**) conventional YOLOv8 detection, (**b**) YOLOv8 combined with SAM for rotated bounding boxes, and (**c**) visualization of the results of YOLOv8-OBB localization

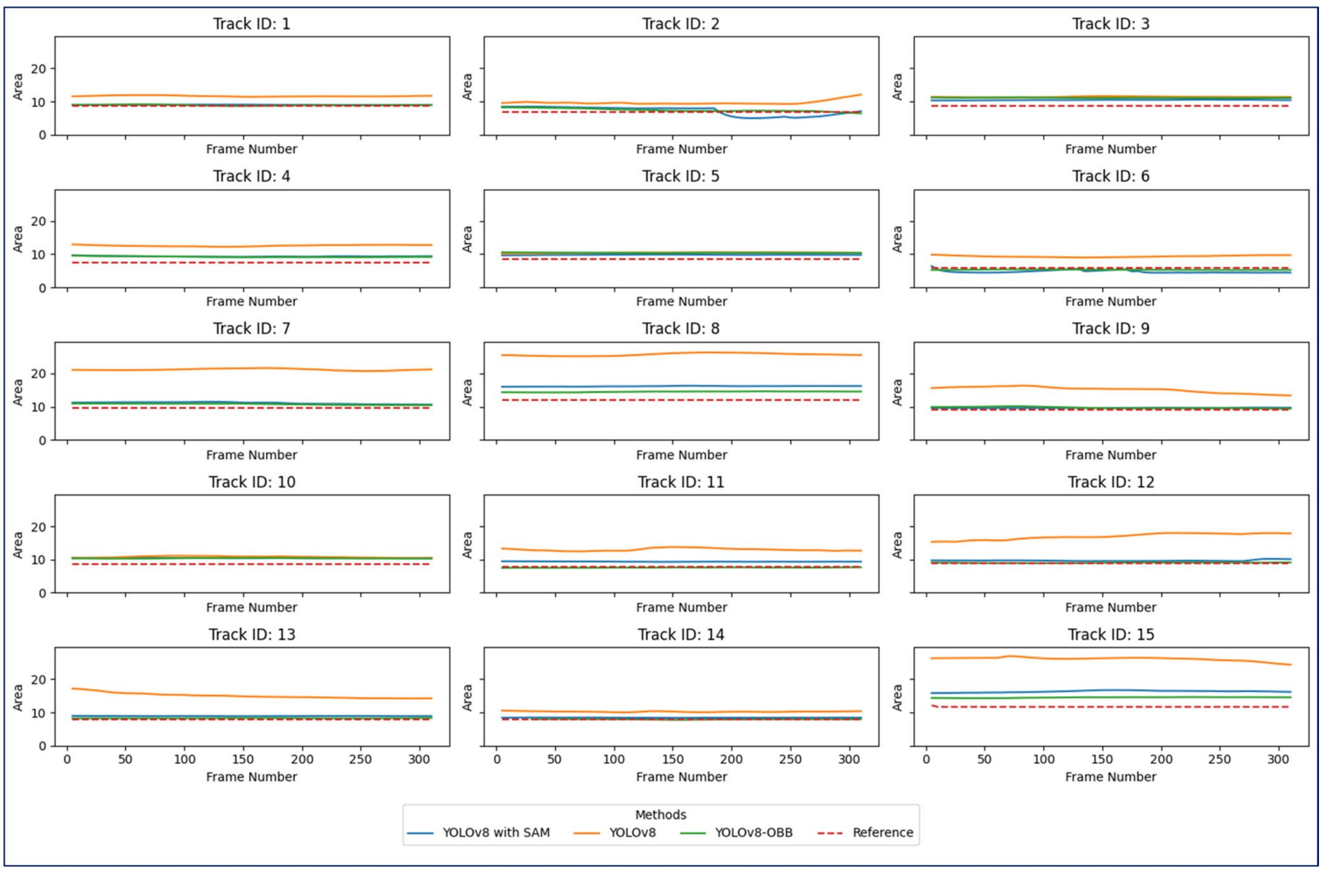

Fig. 7 Area comparison of detected objects (in m^2) from the first footage across sample frames

object. Since each method localizes objects differently, each produced different bounding box dimensions for the exact vehicle (see Figs. 6 and 7). The extracted bounding boxes were used to estimate the vehicle footprint in square meters and compared with the manually obtained reference measurements from QGIS.

Comparing the dimensions of geolocalized vehicles with the reference values revealed that the dimensions estimated from YOLOv8 showed the highest deviation from the reference values (see Fig. 7).

A certain level of variability is produced by the geolocalization workflow, and quantifying this variability is essential for ascertaining each model's effectiveness for road safety applications. As previously emphasized, overestimated bounding boxes can result in false positives for TTC or headway estimations. The geolocalized bounding boxes were further examined by calculating the Root Mean Square Error (RMSE) for each track, aggregated to assess the variability generated by each algorithm.

For each track *(t)*, RMSE was calculated by Eq. (1):

Table 2 Comparison of each model's internal variabilities based on aggregated RMSE

Algorithm	Mean (μ)	Standard Deviation	CI (95%) Lower	CI (95%) Upper
YOLOv8	3.31	2.41	2.87	3.82
YOLOv8 + SAM	1.32	0.87	1.13	1.52
YOLOv8-OBB	0.99	0.93	0.80	1.20

$$RMSE_t = \sqrt{\frac{1}{n_t}\sum_{i=1}^{n_t}(\Delta_{i,t})^2} \quad (1)$$

Where:

$RMSE_t$ is the root mean square error for each track t in m^2, n_t is the number of frames for each track. $\Delta_{i,t}$ is the error for the *i-th* frame of the track representing the difference between the estimated vehicle dimensions from the algorithm and the reference value or actual area. $\frac{1}{n_t}$ is the average of squared errors.

The analysis, discussed in Table 2 and illustrated in Fig. 8, YOLOv8-OBB exhibits the best performance with the lowest mean of 0.99 and a consistent performance (SD = 0.93), with a CI of [0.80, 1.20], suggesting the highest reliability. YOLOv8 + SAM follows with a mean of 1.32, an SD of 0.87, and a CI of [1.13, 1.52], suggesting moderate error and low variability but a performance comparable to YOLOv8-OBB. On the other hand, YOLOv8 shows the highest mean of 3.31, with significant variability (SD = 2.41) and a wide CI of [2.87, 3.82], indicating less consistency and poorer overall performance than its counterparts.

The initial analysis indicates that both YOLOv8-OBB and YOLOv8 + SAM demonstrate comparable accuracy in geolocalization and are suitable for road safety applications within RTM systems. These two algorithms are particularly valuable for identifying critical situations such as Time to Collision (TTC), where underestimation may pose dangers, while overestimation could result in false positive alerts.

Impact of UAV altitude and vehicle rotation on localization

Given that this study involves UAV-based aerial footage, assessing how different factors could impact geolocalization accuracy is crucial. Excessively high altitudes can impair detection precision, as smaller objects may become undetectable. Additionally, the traditional YOLO model relies on axis-aligned bounding boxes (AABB), which usually over-approximate the bounding box of objects that are not axis aligned in the frame or moving objects (see shown in Fig. 9). This issue is alleviated either by redefining the bounding box dimensions through segmentation contours or by introducing a rotation angle from OBB. In the context of road safety analysis and conflict indicators, this oversight could result in false positive alerts in TTC and PET estimations.

$$t = \frac{\left(\bar{X}_1 - \bar{X}_2\right)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_1}}} \quad (2)$$

Welch's t-test was conducted to investigate sensitivity to these two conditions (Eq. 2). It indicates how significantly the means of the groups differ from one another relative to the variation observed in the sample data. Vehicle tracks from the sampled frames were categorized into two independent groups based on the UAV altitude and the vehicle's rotation (see Tables 3 and 4). Welch's t-test was chosen because it accounts for unequal variances between groups, providing a robust assessment of whether differences in RMSE are statistically significant or due to random variation. This approach makes the impact of altitude and vehicle

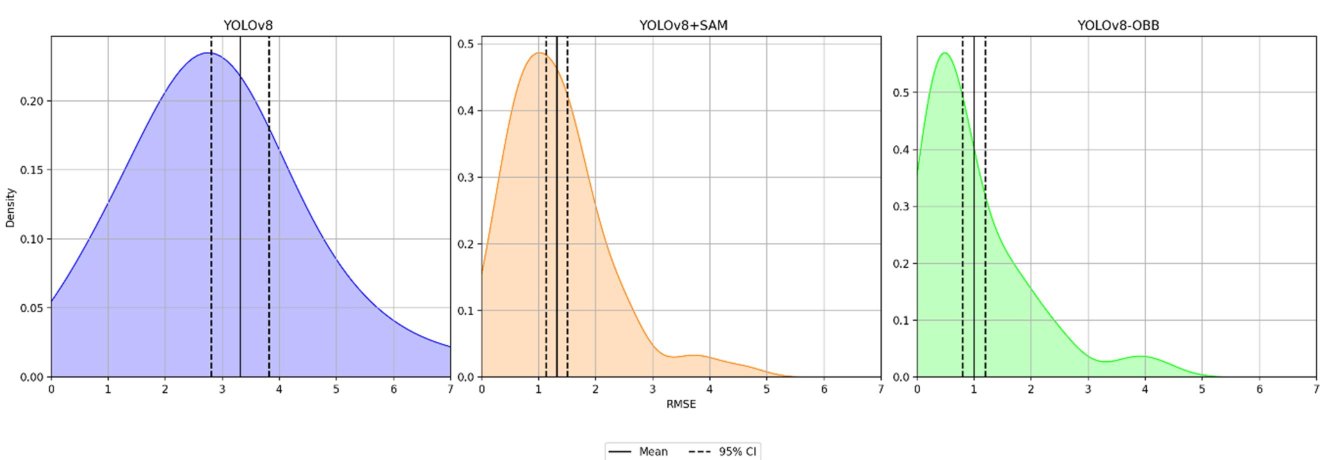

Fig. 8 Kernel Density Estimation (KDE) Plots of Vehicle Dimension Estimations for internal variabilities of each method

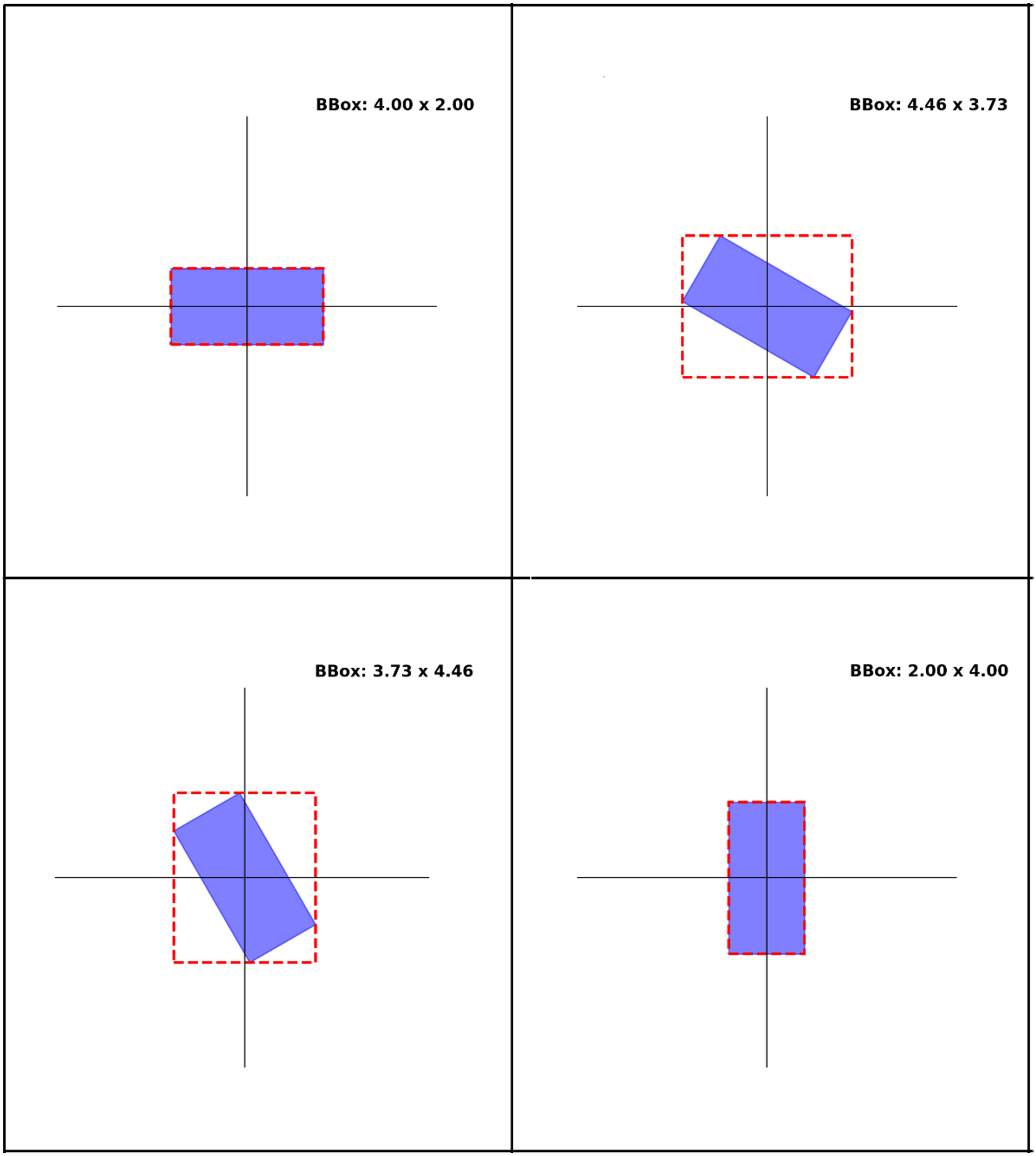

Fig. 9 The rotation angle and its impact on the size of traditional or axis-aligned bounding boxes

orientation more interpretable, with higher absolute t-statistic values indicating greater separation between group means, and values near zero suggesting minimal differences.

Where:

$\bar{X}_1$ and $\bar{X}_2$ are the sample means of two groups, altitudes (60 and 90 m respectively) and rotation (no vehicle rotation and rotated vehicles), s_1^2 and s_2^2 are sample variances, and n_1 and n_2 are sample sizes of the two groups. The p-value indicates the probability that any observed differences occurred by random chance under the null hypothesis.

Table 3 Welch's t-test for the impact of altitude on RMSE for each algorithm

Algorithm	$\bar{X}_1$	$\bar{X}_2$	t-statistic[1]	p-value[2]
YOLOv8	3.226	2.76	0.94	0.37
YOLOv8 + SAM	1.265	1.265	−0.002	0.99
YOLOv8-OBB	0.97	1.09	0.34	0.74

[1](H_0) UAV altitude does not impact the localization accuracy (RMSE) EndFragment

[2]A p-value of less than 0.05 indicates statistically significant differences EndFragment

Table 4 Welch's t-test for the impact of rotation on RMSE for each algorithm

Algorithm	$\bar{X}_1$	$\bar{X}_2$	t-statistic[3]	p-value[4]
YOLOv8	2.80	3.51	−5.07	0.0036
YOLOv8 + SAM	1.26	2.06	−1.08	0.32
YOLOv8-OBB	0.98	0.77	−0.51	0.62

[3] (H_0) = Vehicle rotation does not impact the localization accuracy (RMSE)

[4] A p-value of less than 0.05 indicates statistically significant differences

A p-value below the conventional threshold of 0.05 denotes that the findings are statistically significant.

The results in Table 3 show no significant differences in RMSE of vehicle dimensions at the two altitudes for all three models. The YOLOv8 model had a t-statistic of 0.94 and a p-value of 0.37; the YOLOv8 + SAM model showed a t-statistic of −0.002 and a p-value of 0.99, and the YOLOv8-OBB model had a t-statistic of 0.34 and a p-value of 0.74, supporting the other models' findings. These findings confirm the null hypothesis (H_0) that altitude does not influence the accuracy of bounding box coordinates.

In contrast, Table 4 shows that vehicle rotation impacts the base YOLOv8 model. The traditional YOLOv8 had a t-statistic of −5.07 and a p-value of 0.0036, indicating a statistically significant effect of rotation on bounding box estimates (as illustrated in Fig. 9). However, YOLOv8 + SAM and YOLOv8-OBB showed no significant differences between the two orientations, with t-statistics of −1.08 ($p = 0.32$) and −0.51 ($p = 0.62$), respectively. The results suggest that while the traditional YOLO algorithm with AABB is sensitive to the object's orientation, both segmentation and OBB-enabled configuration maintain consistent performance as these bounding boxes perfectly encapsulate the detected objects, capturing their true dimensions. This makes these algorithms more suited for conflict analysis.

In terms of precision, all tests indicate that YOLOv8 + SAM and YOLOv8-OBB yield comparable results, demonstrating minimal error and variability. Additionally, these models are resistant to the effects of vehicle orientation, a limitation posed by the standard YOLOv8 model. However, a notable drawback is the computational expense associated with segmentation. When comparing the three algorithms in speed, YOLOv8 is the fastest, while YOLOv8-OBB operates at 1.5 times slower speed. The slowest of the three is YOLOv8 + SAM, which is 382 times slower than YOLOv8. While its performance can be enhanced using a GPU instead of a CPU, this still limits the feasibility of applications requiring near-real-time localization, which is critical in RTM systems. This experiment confirms that OBB-enabled variants of YOLO are particularly best-suited for road safety applications and can effectively localize vehicles for road safety analysis.

Conclusion

This study presents a vehicle geolocalization method from UAV data that combines state-of-the-art DL-based object detection and segmentation methods with a feature-matching algorithm to geolocalize road users. The focus was on the dimensional accuracy, as the study was aimed at identifying a technically viable solution for road safety applications. Three different combinations of YOLOv8 capable of object detection, segmentation, and OBB were used.

Each algorithm was tested rigorously to evaluate the best algorithmic configuration for this use case. The evaluation included the overall precision of each method, the impact of the UAV's altitude and vehicle's orientation on geolocalization accuracy, and the inference time. The experimental results reveal that integrating YOLOv8 with SAM and using the OBB-enabled YOLO model notably improved the system's ability to accurately detect and localize vehicles in real space and preserve their shapes. However, the inference time of segmentation is steep; therefore, near-real-time and real-time implementation is practically infeasible. Hence, OBB-enabled YOLO models are best suited for road safety applications. The study's findings will have profound implications in this domain, as it is a significant step towards automation in road safety, where dimensional accuracy is paramount. Despite these benefits, the study also acknowledges the limitations of visual-based methods. Limitations include dependency on clear-sky and daylight conditions and UAV battery constraints, which restrict prolonged data collection. Nevertheless, the advantages of the aerial perspective and AI integration outweigh these drawbacks, making UAVs an effective tool for traffic monitoring and safety analysis.

In conclusion, the study presents a strong argument for adopting UAV technology and AI for road safety systems. These disruptive technologies can offer scalable and effective solutions for vehicle detection and geolocalization challenges, paving the way for the next generation of intelligent transportation systems. As we move forward, further research and development must continue to build upon these foundations, exploring new possibilities and pushing the limits of what can be achieved in road traffic safety and management.

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Data availability The data presented in this study are available on request from the corresponding author due to privacy concerns.

Declarations

Competing interests The authors declare no competing interests.

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